Spectrophotometer laboratory report

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This laboratory report presents a series of spectrophotometry analysis of the absorption and transmittance properties of various samples using a Perkin-Elmer Lambda 9 spectrophotometer. The study determines the concentration of a liquid dye solution with unknown concentration to be 1.2936%. Another analysis showed that acrylic, normal glass, and polycarbonate exhibit 0% transmittance at 254 nm; however, at 365 nm, only acrylic remains opaque while the other two materials allow significant light transmission of over 80%. Analysis of optical properties of Polished silicon and green polypropylene revealed that the duo blocked visible and near-IR spectrum up to about 800 nm but become transparent with transmittance values of about 35% for silicon and approximately 1% for green PP at 1490 nm, making silicon preferable for IR sensor casings. Material-specific absorptions were also identified at 2239 nm for acrylic, 3200 nm for polycarbonate, and 839 nm for green PP, enabling their identification in IR region. Laser safety glasses showed effective absorption at 308 nm and 633 nm, showing their protective performance against xenon chloride and helium-neon laser sources.

I.

A. Theory

spectrophotometer has wide spread adoption across numerous scientific disciplines, including biology, chemistry, and materials science. This is due to the broad range of sample types at which it can analyze, coupled with the instrument's user-friendly operation.

When electromagnetic radiation passes through a sample, some wavelengths are absorbed by the constituents materials, while others are transmitted through the medium[1].

Consequently, spectrophotometers are designed to quantify optical properties such as reflectance, absorbance and transmittance of a material.

Spectrophotometer has wider range of applications including analysis of water quality[2] solution's properties characterization [3]. Spectrophotometer system is also utilized in the analysis of concentration of chemical compounds in a liquid due to its sensitivity and non-destructiveness[4]. The instrument is also employed in the measurements of collision broadening and shift in optical absorption spectra [5].

Furthermore, spectrophotometers are employed to measure optical transmittance of solids and liquids and also the molar absorptivity of different chemical samples, which are fundamental parameters of interest in characterizing materials [6].

Fourier transform infrared spectrophotometer can also be used to quantify diffuse reflectance (DR) from in addition to the normal transmittance measurements[7].

B. Beer Lambert law

This Law describes the connection between a sample's light absorbance and the concentration of the material, provides the foundation for how a spectrophotometer operates.

In effect, one can determine the concentration of a specific chemical sample in a solution when the sample's absorption at a particular wavelength is known [8]. The relationship between the absorption and concentration is give by the Beer Lambert law as equation 1

$$A = aCl \tag{1}$$

where A is the absorption, a is the molar concentration and C is the concentration and l is the optical path-length of the sample [9].

Furthermore, According to Lambert Beer's law, when light traversing a material, the ratio of transmitted light intensity I and the incident light intensity I_0 is defined as transmittance T. This ratio decreases exponentially as the light traversing a sample, in accordance with the equation:

$$T = \frac{I}{I_0} = \exp(-\alpha l), \tag{2}$$

where α is the absorption coefficient, which is material's dependent. The absorbance A is a dimensionless empirical quantity defined in equation 1, and also as the negative logarithm of the transmittance, that is, $A = -\log T$. Thus, we can combine equation 1 and 2 as

$$A = \log\left(\frac{I_0}{I}\right) = aCl, \tag{3}$$

The equations play a vital role in determining the unknown concentration of a given sample.

II.

A. Materials and methods

In the experiment, a spectrophotometer Perkin-Elmer Lambda 9 given in figure 1

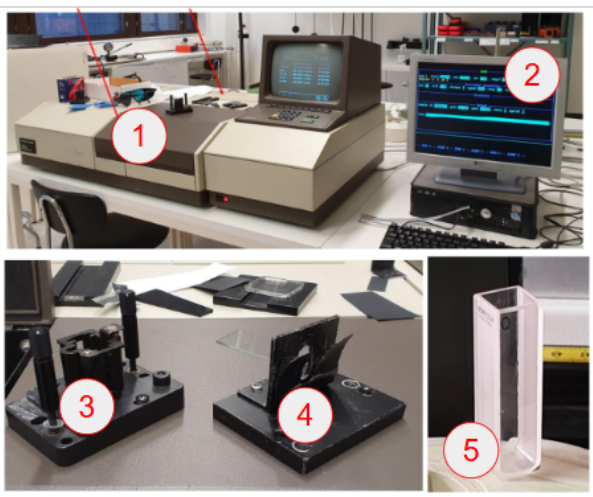


FIG. 1. Perkin-Elmer Lambda 9 spectrophotometer; 1. Sample compartment, 2. Software interface, 3. Liquid sample holder, 4. Solid sample holder, and 5. Quartz cuvette for measuring fluids[10]

was employed to measure both absorption in liquids and transmittance in solids. Two sets of samples(liquid and solid) were provided for analysis.
The liquid samples provided for the analysis comprised a reference solution with 0% colorant concentration, and other solutions of 0.5%, 1.0%, 1.5%, and 2.0% colorant concentration, and an additional sample with an unknown concentration.
The solid samples provided for the analysis included a standard microscope glass slide, acrylic (polymethyl methacrylate PMMA), polycarbonate (PC), a polished silicon wafer, green polypropylene (PP), a pair of laser safety glasses and an unknown glass sample (the spectrophotometer cuvette).
To measure the absorption by the liquid samples, the experiment began with the following calibrations in the software interface: Start =800 nm, End= 200 nm, Ord = A, Speed =240 nm/min , Ymin =0, Ymax =4.
The spectrophotometer cuvette was filled with a liquid sample, and was placed in the liquid holder in the spectrophotometer sample compartment, and the reading was done by using the software interface, and the results were saved for all the different concentrations.
In a similar way, to measure the transmittance through the solid samples, the calibration of the spectrophotometer was done with the following parameters, start 3200 nm, End 200 nm, Ord = T, Speed 960 nm/min, and the graphics automatically scaled right (0- 100 % transmittance).The solid sample was placed in the solid sample holder in the spectrophotometer sample compartment, and the results were saved for all the different samples.

III.

A. Results

This section is dedicated for the analysis of the results obtained from the spectrophotometer according the given tasks.

B. Measurement of unknown concentration

Absorption and transmittance of electromagnetic radiation through materials are wavelength dependent. This property has made it possible for one to study the material transmission and absorption for different wavelengths. To obtain the concentration of the fluid, we plot the wavelength and the absorption as shown in figure 2.

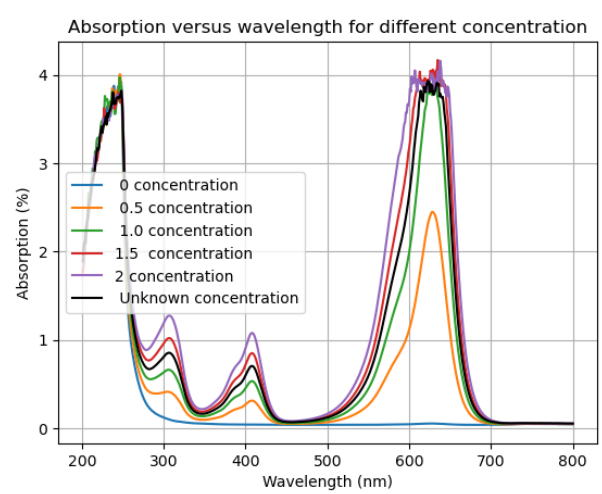
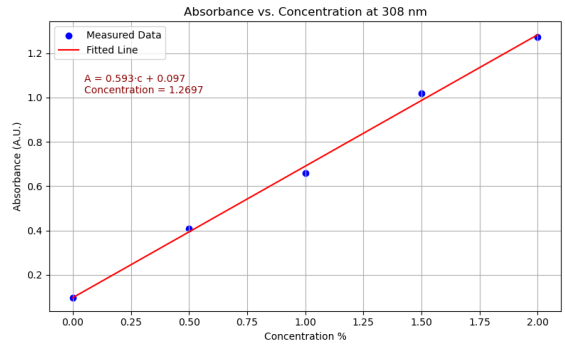


FIG. 2. Absorption versus wavelength

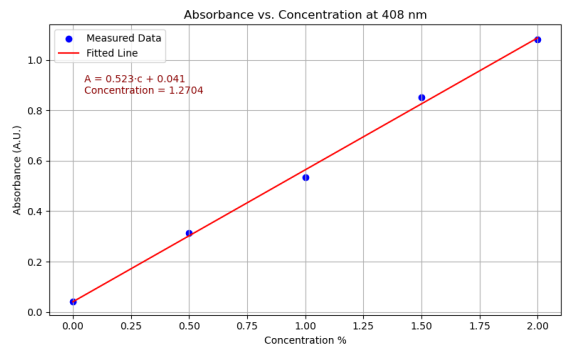
From the graph it is clearly seen that absorptions

occurs at four distinct wavelengths region, approximately, 250 nm, 300 nm 400 nm and 600 nm. Ammong these, the peak absorptions at the region of 300 nm and 400 nm were computed numerically to be 308 nm, 408 nm respectively. In addition, at 600 nm, the absorbance also varies with the liquid concentrations. The optimal wavelength for calculating absorbance based on concentration was defined to be 408 nm, where the reference liquid with zero concentration shows zero absorbance, and the absorbance peaks of other concentrations were clearly distinguishable without any saturation. The blue line is the reference liquid with zero concentration, and the black line is the liquid with unknown concentration. The red curve is the liquid with a concentration of 1.5%, and the green line is the liquid with the concentration of 1% . The orange curve is the concentration of 0.5%, and the purple is the concentration of 2%. The absorbance of the unknown concentration lies between the 1% and 1.5% concentrations. Moreover, to be more, plot fittings for the concentration and the absorption for these three wavelength, 308nm, 408nm and 600nm using 1 is presented in the figure [2] a to c respectively.

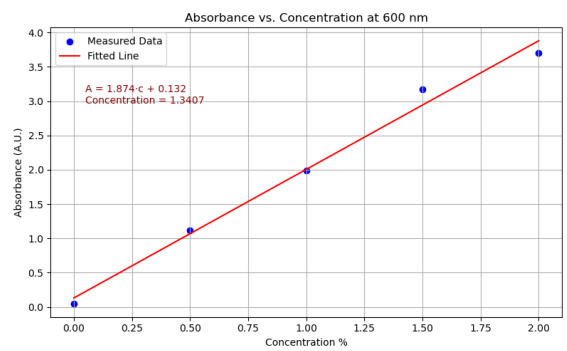
In each of the wavelength, the graphs show linear relationship. The absorption of the unknown concentration were found to be 0.8503, 0.7054,and 2.6447% for the wavelength of 308nm, 408nm and 600nm respectively. From the fitting line, the values of the unknown concentration was obtained to be 1.2697,1.2704 and 1.3407 for 308nm, 408nm and 600nm respectively.The average of the unknown concentration is found to be 1.2936%.



(a) Fitting line for 308 nm



(b) Fitting line for 408 nm



(c) Fitting line for 600 nm

FIG. 3. Line fitting for concentration

C. Finding material for UV-protection

It is well known the effect of ultraviolet radiation on humans and other organisms. Thus, finding materials that block this radiation is a crucial research area of interest. In this section, several materials' transmittance across different wavelengths are studied and the graph of the transmittance against the wavelength is presented in figure 4

The task here is to determine the material that would block UV radiation from a mercury lamp among polycarbonate, acrylic and normal glass.

For proper visualization, the graph for the materials(Normal glass, Acrylic and Polycarbonate) provided for this analysis is presented in the figure 5. The red dotted line represents the wavelength (254 nm) that the mercury lamp emits UV radiation.

The transmittance spectra showed that the acrylic had negligible transmittance just above 250 nm, whilst polycarbonate and standard glass began to transmit light noticeably at approximately 290 nm. In addition, all tested materials exhibit high transmittance in the visible regime (wavelengths = 380 nm), indicating optical transparency in this region. The optical properties of the materials described above suggest that they are well-suited for applications where UV radiation at 254 nm (the characteristic emission line of mercury vapor lamps) needs to be effectively blocked, while allowing full transmission of visible light.

Additionally, the optical properties of the materials was also analyzed at 365 nm. These results showed that only acrylic absorbs relatively strong at this wavelength making it a good candidate when appli-

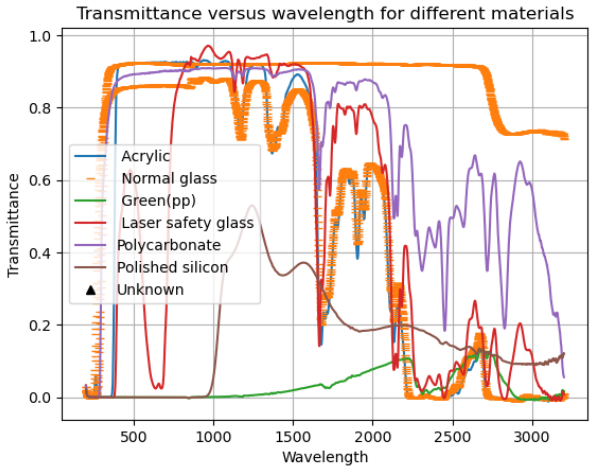


FIG. 4. Transmittance versus wavelength for different materials

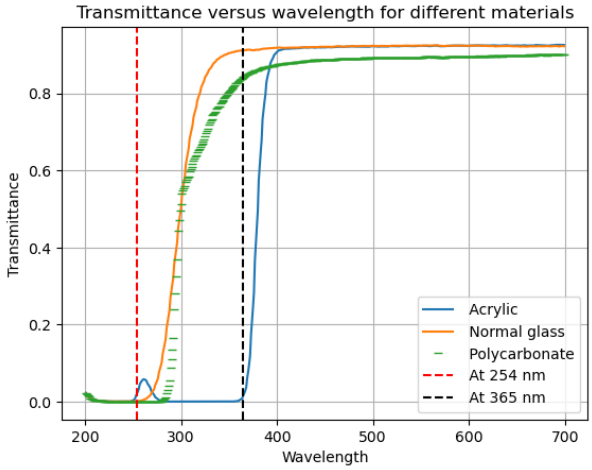


FIG. 5. Transmittance for materials for uv protection

cations need blocking of the given wavelength. On the other hand, high transmittance up to this wavelength can be observed for polycarbonate and normal glass, so both remain transparent in the visible region.

D. Finding material to block visible light, but transmitting IR

The visible spectrum ranges from 380 nm to 700 nm. Two materials (green pp and polished silicon) were provided for investigation to block this wavelength range but transmit at infrared range (above 800 nm). Figure 6 shows the graph of the transmittance versus wavelength of the materials.
From the graph, it can be deduced that the materials are opaque in the visible wavelength region but transparent in the infrared region. Therefore, the two materials—polished silicon disc and green polypropylene (PP) are excellent at blocking visible light, while also being effective in transmitting infrared light. Moreover, for better detector performance, polished silicon is best suited due to its higher transmittance of about 35% at 1450 nm compared to green PP which has the transmittance of about 1%.

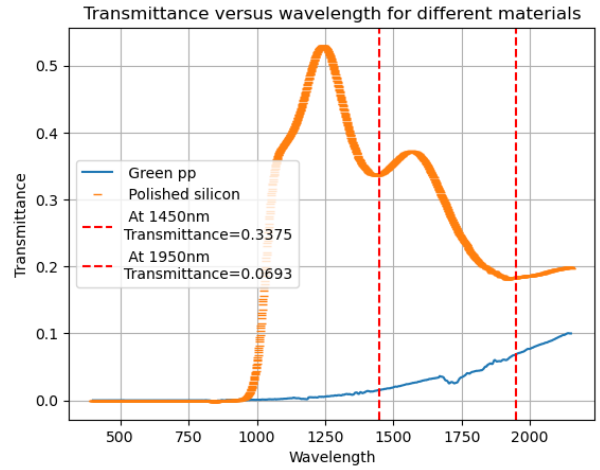


FIG. 6. Transmittance of polished silicon and green pp in IR region

E. Recognizing type of plastic

In the infrared range, the absorption by materials molecules correlates with the molecular structure of materials, as distinct materials possess varying molecule configurations. This makes it possible to recognize material by finding absorption wavelengths. A careful observation from the figure 7, reveals that acrylic, polycarbonate and polypropylene have their lowest transmittance (corresponding to maximum absorption) at wavelengths 2239 nm, 3200 nm and 839 nm respectively. Therefore the materials can easily be identified with these wavelength.

F. Laser safety glasses

Laser safety glasses were provided to identify the laser source for which the glasses can offer protection against the laser light from entering the eyes. The plot for the transmittance against wavelength is given in figure 8.

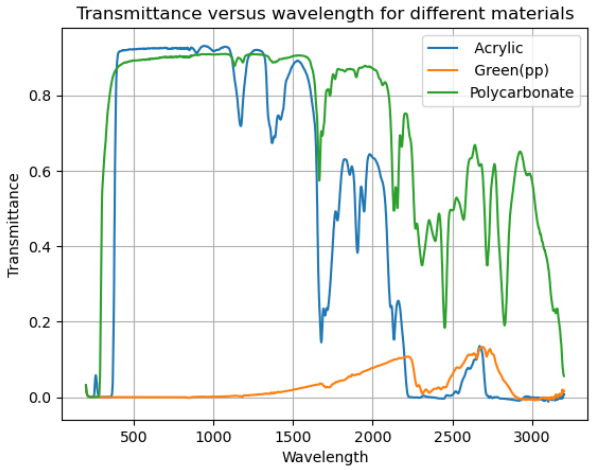


FIG. 7. Transimttance against wavelength for different materials

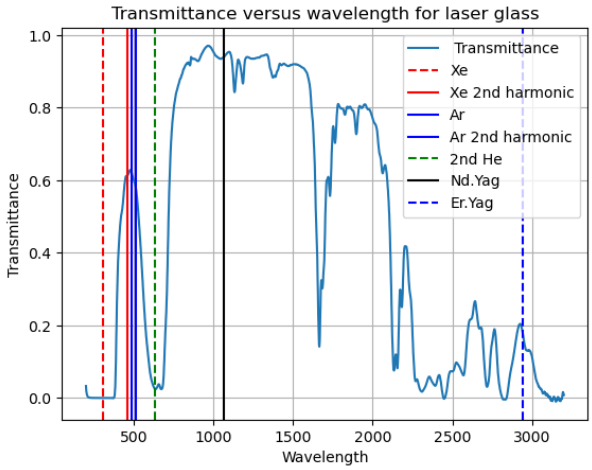


FIG. 8. Transmittance versus wavelength of laser safety glass

From the graph, the vertical lines represent different wavelengths for different laser sources. The red dotted lines represent XeCl laser with the wavelengths of 308 nm and transmission of about 0%, the green dotted lines represent second harmonic He with the wavelengths of 633 nm and transmission of about 1%, and the blue dotted lines represent Er Yag with the wavelengths of 2940 nm and transmission of about 20%. The laser safety glasses were designed to protect the user from the wavelength of HeNe laser and Xenon first harmonic. However, the laser safety glasses is not protective for the XeCl with wavelength 459 nm and transmittance of about 60%, Er Yag with wavelength of 2940 nm and 20% transmittance, Argon laser of wavelength 488 and 514 nm having transmission of about 60% for both wavelengths, Ruby laser of wavelength 694 nm with the transmission of about 60%, and Nd:Yag laser with the wavelength 1064 nm and the transmission of about 90%.

IV. CONCLUSION

This study successfully demonstrated the applicability of Perkin-Elmer Lambda 9 spectrophotometer to characterize the optical absorption and transmittance properties of a range of liquid and solid materials across ultraviolet, visible, and infrared spectral regions. The concentration of an unknown dye solution was accurately determined to be 1.2936% through spectrophotometry analysis.

Optical evaluation of acrylic, standard glass, and polycarbonate revealed that all three materials exhibit complete opacity (0% transmittance) at 254 nm, aligning with the mercury lamp emission. At 365 nm, standard glass and polycarbonate displayed high transmittance, while acrylic remained opaque, indicating selective UV transparency among these materials.

Polished silicon and green polypropylene were shown to be opaque up to approximately 800 nm but became increasingly transparent in the infrared region, particularly around 1000 nm. At 1490 nm, polished silicon demonstrated superior transmittance (35%) compared to green PP (1%), making it the preferred candidate for infrared sensor housings requiring visible light blocking and IR transparency.

Material-specific absorption features in the IR range were also identified, with acrylic, polycarbonate, and green PP exhibiting strong absorptivity at 2239 nm, 3200 nm, and 839 nm respectively, facilitating their spectral identification.

Lastly, laser safety glasses was confirmed to provide effective protection, achieving nearly 0% transmittance at 308 nm (XeCl excimer laser) and approximately 1% at 633 nm (He-Ne laser), verifying their suitability for dual-wavelength laser shielding applications. However, the laser safety glasses is not protective for the Er Yag with 20% transmittance, Argon laser of wavelength 488 and 514 nm having transmission of about 60% for both wavelengths, Ruby laser of wavelength 694 nm with the transmission of about 60%, and Nd:Yag laser with the wavelength 1064 nm and the transmission of about 90

[1] A. Mishra, R. Kumar, S. N. Mishra, S. Vijayaraghavalu, G. C. Shukla, and M. Kumar, The effects of psychological and biological components in tobacco consumption in india, Available at SSRN 4208545.

[2] J. Suwanboriboon, W. Meesiri, and W. Wongkokua, An application of spectrophotometer for admi color measurement, in *Journal of Physics: Conference Series*, Vol. 1144 (IOP Publishing, 2018) p. 012064.

[3] B. J. Russell, J. P. Shelton, and A. Walsh, An atomic-absorption spectrophotometer and its application to the analysis of solutions, Spectrochimica Acta **8**, 317 (1957).

[4] P. Listiaji and G. B. Suparta, Low-cost imaging spectrophotometer system for absorbance measurement, Journal of Physics: Conference Series **1567**, 042093 (2020).

[5] A. Ghosh, An automatic rapid scanning spectrophotometer to study thermoluminescence spectra, Applied Optics **2**, 243 (1963).

[6] R. Mavrodineanu, An accurate spectrophotometer for measuring the transmittance of solid and liquid materials, Journal of Research of the National Bureau of Standards. Section A, Physics and Chemistry **76**, 405 (1972).

[7] R. R. Willey, Fourier transform infrared spectrophotometer for transmittance and diffuse reflectance measurements, Appl. Spectrosc. **30**, 593 (1976).

[8] J. H. Hardesty and B. Attili, Spectrophotometry and the beer-lambert law: An important analytical technique in chemistry, Collin College, Department of Chemistry , 1 (2010).

[9] F. A. de Melo Marques, O. J. de Rezende Silveira, I. F. de Moura Domingues, D. C. Fuzatto, A. A. C. Cotta, J. E. Tsuchida, and J. H. Osório, Instrumentation applied to the teaching of optics and photonics: revitalizing and automating a spectrophotometer, Physics Education **60**, 035021 (2025).

[10] Lecture material, spectrophotometer measurements, , 1 (2024/2025).